Transformations & Symmetry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

2 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 12. Substituting $x = 1$ into $y = 4x + 8$ gives $y = 4(1) + 8 = 12$.

3 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 94°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 52^\circ) = 94^\circ$.

4 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

5 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 16. Substituting $x = 2$ into $y = 4x + 8$ gives $y = 4(2) + 8 = 16$.

6 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 87°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 59^\circ) = 87^\circ$.

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7 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 32 cm. Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 20. Substituting $x = 3$ into $y = 4x + 8$ gives $y = 4(3) + 8 = 20$.

9 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 80°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 66^\circ) = 80^\circ$.

10 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 40 cm. Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

11 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 24. Substituting $x = 4$ into $y = 4x + 8$ gives $y = 4(4) + 8 = 24$.

12 [Intermediate] • Target: Properties of trapezoids

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

13 [Intermediate] • Target: Properties of trapezoids

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

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14 [Intermediate] • Target: Properties of trapezoids

A data set recorded for properties of trapezoids consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 3.01 Triangles & Congruence

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(3)$.

Final Answer: 15. Substitute $x = 3$: $f(3) = 2(3) + 9 = 15$.

16 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

17 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 5.01 Right Triangles & Trigonometry

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(4)$.

Final Answer: 17. Substitute $x = 4$: $f(4) = 2(4) + 9 = 17$.

18 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

19 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 7.01 Circles

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(5)$.

Final Answer: 19. Substitute $x = 5$: $f(5) = 2(5) + 9 = 19$.

20 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 8.01 Area, Surface Area & Volume

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 103° ✓

b. 113°

c. 95°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

22 [Foundational] • Target: Angles of polygons

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

23 [Foundational] • Target: Angles of polygons

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 10

b. 14

c. 12 ✓

d. 14

Substituting $x = 3$ into $y = 2x + 6$ gives $y = 2(3) + 6 = 12$.

24 [Foundational] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 82° ✓

b. 92°

c. 74°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 66^\circ) = 82^\circ$.

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25 [Foundational] • Target: Properties of parallelograms

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

26 [Foundational] • Target: Properties of parallelograms

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

27 [Foundational] • Target: Properties of parallelograms

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 13

b. 18

c. 16 ✓

d. 19

Substituting $x = 3$ into $y = 3x + 7$ gives $y = 3(3) + 7 = 16$.

28 [Foundational] • Target: Properties of parallelograms

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 66^\circ) = 81^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Angles of polygons

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

30 [Intermediate] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96° ✓

b. 106°

c. 88°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 52^\circ) = 96^\circ$.

31 [Intermediate] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 89° ✓

b. 99°

c. 81°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 59^\circ) = 89^\circ$.

32 [Intermediate] • Target: Angles of polygons

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

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33 [Intermediate] • Target: Properties of parallelograms

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

34 [Intermediate] • Target: Properties of parallelograms

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 52^\circ) = 95^\circ$.

35 [Intermediate] • Target: Properties of parallelograms

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

36 [Intermediate] • Target: Properties of parallelograms

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 16

b. 21

c. 19 ✓

d. 22

Substituting $x = 4$ into $y = 3x + 7$ gives $y = 3(4) + 7 = 19$.

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37 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 14

c. 12 ✓

d. 16

Substituting $x = 1$ into $y = 4x + 8$ gives $y = 4(1) + 8 = 12$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Properties of rectangles, rhombuses, and squares • Cross-Lesson: 6.01 Polygons & Quadrilaterals

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 12

b. 18

c. 16 ✓

d. 20

Substituting $x = 2$ into $y = 4x + 8$ gives $y = 4(2) + 8 = 16$.

39 [Challenging] • Target: Properties of rectangles, rhombuses, and squares • Cross-Lesson: 7.01 Circles

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

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40 [Challenging] • Target: Properties of rectangles, rhombuses, and squares • Cross-Lesson: 8.01 Area, Surface Area & Volume

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24 ✓

d. 28

Substituting $x = 4$ into $y = 4x + 8$ gives $y = 4(4) + 8 = 24$.

41 [Challenging] • Target: Classifying quadrilaterals • Cross-Lesson: 1.01 Foundations & Reasoning

A data set recorded for classifying quadrilaterals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

42 [Challenging] • Target: Classifying quadrilaterals • Cross-Lesson: 10.01 Coordinate Geometry

For the model $f(x) = 3x + 3$ in classifying quadrilaterals, evaluate $f(4)$.

a. 15 ✓

b. 18

c. 12

d. 19

Substitute $x = 4$: $f(4) = 3(4) + 3 = 15$.

43 [Challenging] • Target: Classifying quadrilaterals • Cross-Lesson: 11.01 Probability

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

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44 [Challenging] • Target: Coordinate methods with quadrilaterals • Cross-Lesson: 2.01 Lines & Angles

A data set recorded for coordinate methods with quadrilaterals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

45 [Challenging] • Target: Coordinate methods with quadrilaterals • Cross-Lesson: 3.01 Triangles & Congruence

For the model $f(x) = 4x + 4$ in coordinate methods with quadrilaterals, evaluate $f(4)$.

a. 20 ✓

b. 24

c. 16

d. 24

Substitute $x = 4$: $f(4) = 4(4) + 4 = 20$.

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